SDR 184 Final Project: eSports Earning Analysis

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Introduction

eSports is a really interesting topic because of how global it is. Almost anyone in the world can compete and there's competition in almost every game imaginable, and when we saw that the eSports earnings website was one of the data sources available for this project, we were interested.

This data contains the eSports earning for players for the past 20+ years. It contains data from every major to mid sized eSport, from players from pretty much every country in the world. This data was chosen due to it being really rich and having a lot of different aspects to analyze. The way this data was obtained was from https://www.esportsearnings.com/, and all the data in this website was entered manually, getting it from forums, tournament organizers, and all made sure to be accurate.

This data follows the FAIR principles because this data is pretty findable since you can go to that website and find all the data that I used, its accessible since its free and anyone can use it for whatever they need, its interoperable because you can modify it and pretty easily, and it's reusable because the data will stay up and anyone can use it.

This follows the CARE principles because it is for collective benefit since it's cool and interesting and helps people understand how everyone can be good at esports, this data has authority to control because people are empowered by it, and its fair. It's responsible because it has no bias, and it accurately represents groups. It's ethical because it hurts no one and it makes sure to treat everyone equally.

The specific attributes of this data that im focused on, is specifically the geographical differences between players of different games, and what regions excel in different games. There's a lot of data on different players and where they're from, and this could make a really interesting analysis.

First Analysis

The first data analysis I decided to do, was to take the statistics of the top earning countries of all time and analyze it. I took the data from eSports earnings which gives the highest earning countries, with the amount of money, and how many players in each country. I was really curious what countries in general are the best at eSports since it could be anything. Tidying this in R gave me this data:

	Rank	Country	Total Earnings (USD)	Players	Earnings per Player (USD)
1	1	China	$333,124,854	9,387	$35,488
2	2	United States	$302,289,324	29,501	$10,247
3	3	Korea, Republic of	$157,154,935	5,987	$26,249
4	4	Russian Federation	$98,801,881	5,900	$16,746
5	5	Brazil	$74,212,435	6,058	$12,250
6	6	Denmark	$64,982,128	2,212	$29,377
7	7	France	$63,290,937	6,705	$9,439
8	8	Sweden	$59,203,302	3,367	$17,583
9	9	Germany	$53,321,255	6,987	$7,631
10	10	Canada	$50,919,207	4,395	$11,586
11	11	United Kingdom	$50,674,464	5,747	$8,818
12	12	Japan	$43,535,933	4,004	$10,873
13	13	Ukraine	$41,266,588	1,727	$23,895
14	14	Australia	$34,849,292	4,639	$7,512
15	15	Finland	$34,242,303	2,216	$15,452
16	16	Poland	$32,938,128	3,164	$10,410
17	17	Thailand	$31,614,841	2,319	$13,633
18	18	Philippines	$27,381,660	1,918	$14,276
19	19	Malaysia	$23,483,101	1,723	$13,629
20	20	Taiwan, Republic of China	$23,126,488	1,726	$13,399

Using the tidied data from this graph, I was able to create a visualization using a bar graph, which gave me this:

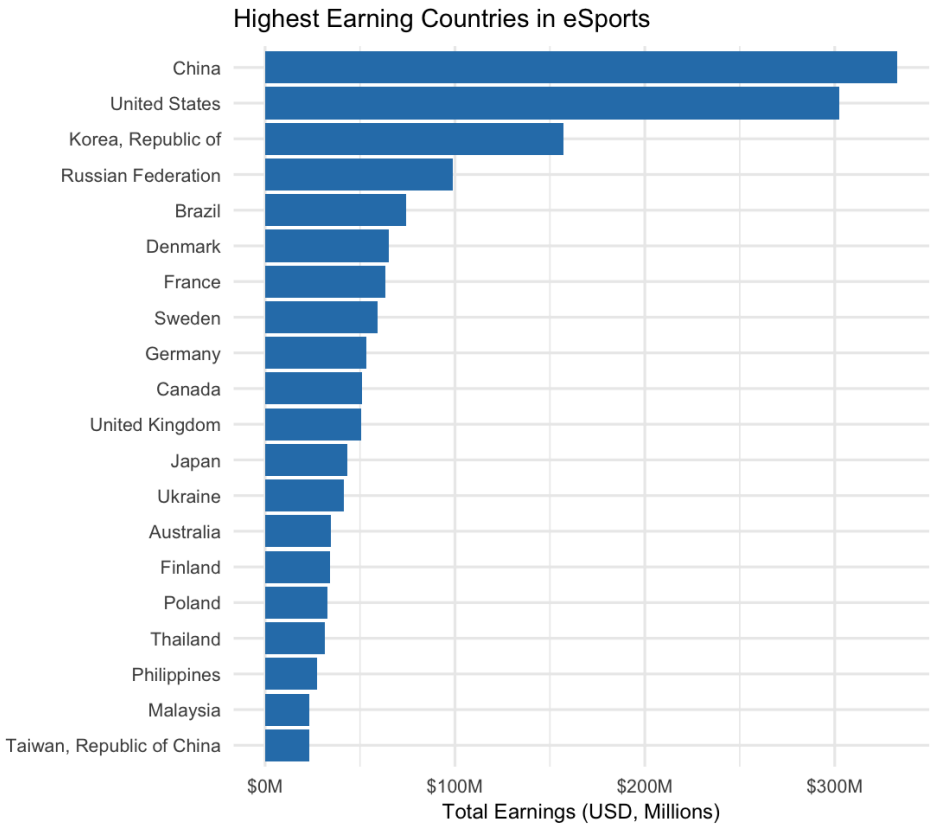
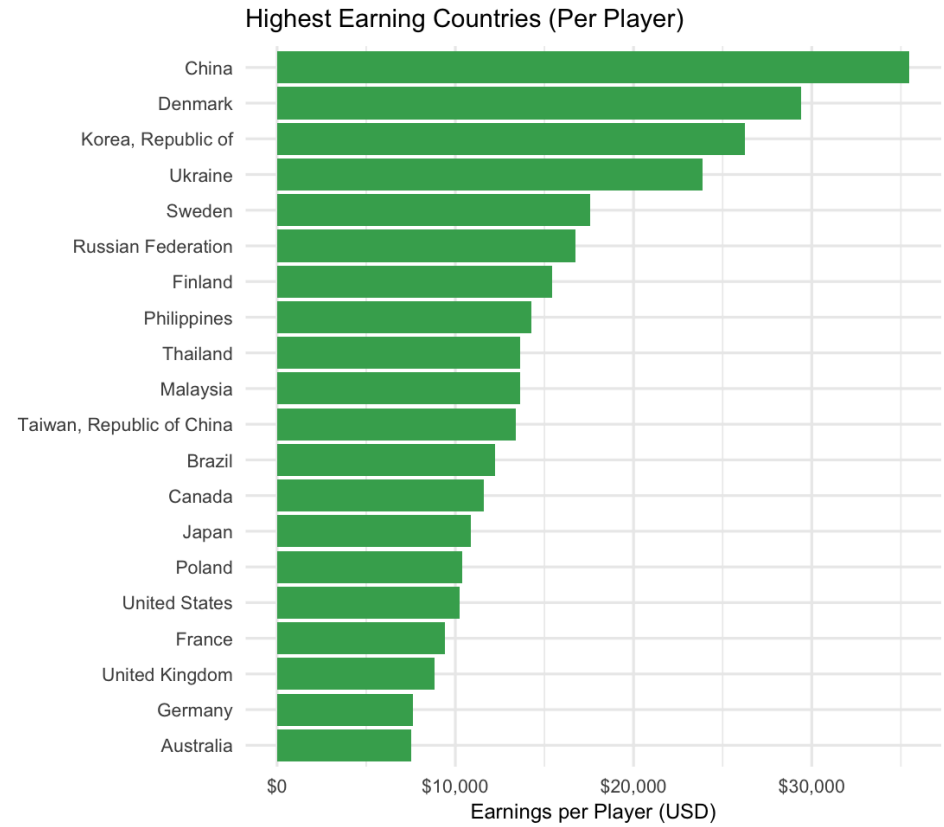


Figure 1. Total eSports earnings by country for the top 20 earning countries (USD, shown in millions).

Alt text: This graph is a horizontal bar graph that demonstrates the highest earning countries in eSports. At the very top, we have China, United States Korea, Russia, and Brazil. Then, it decreases pretty linearly. China has earned around 350 millon USD, and has a pretty exponential decrease.

There's a few conclusions I've reached from this data. Firstly, a country having a bigger population has a direct correlation with how much success a country has in eSports. China, United States, Russia, And Brazil are some of the biggest countries in the world, and this makes sense because if there's more people in a country, there's going to be more people competing.

I also figured because of the bigger countries having more people, I would make a graph that adjusts for the earnings per player in each country so its fairer for each country. After doing this, I ended up with this graph:



er by country (total earnings divided by player count). This adjusts for different numbers of players across countries.

Alt text: This graph represents the highest earning countries per player. At the very top, we have China at $35,000. Then it goes down being Denmark, Korea, Ukraine, Sweden, and then more. As can be seen, the top countries change a decent amount when comparing top countries when divided by the amount of players. Specifically, the European countries tend to jump up quite a bit. They have less players competing overall, but in the games that they do indeed compete in, they tend to be extremely successful in. China is still in the top due to their success in almost every time of game, and the top tends to be dominated by Asia and Europe in almost every type of games.

Second Analysis

The second analysis I've decided to do is a stacked bar chart of the top 10 games per country, looking specifically at the top three countries (China, United States, Korea). I took the data from eSportsearnings, which shows top games per country. I made a separate table and graph for each of the three countries.

	Game	Pct
1	Dota 2	30.156010
2	Arena of Valor	29.155023
3	PLAYERUNKNOWN'S BATTLEGROUNDS Mobile	12.628404
4	League of Legends	8.355541
5	CrossFire	6.597197
6	PLAYERUNKNOWNå□□S BATTLEGROUNDS	5.624853
7	CrossFire Mobile	2.353099
8	League of Legends: Wild Rift	1.809400
9	Naraka: Bladepoint	1.667873
10	Identity V	1.652600

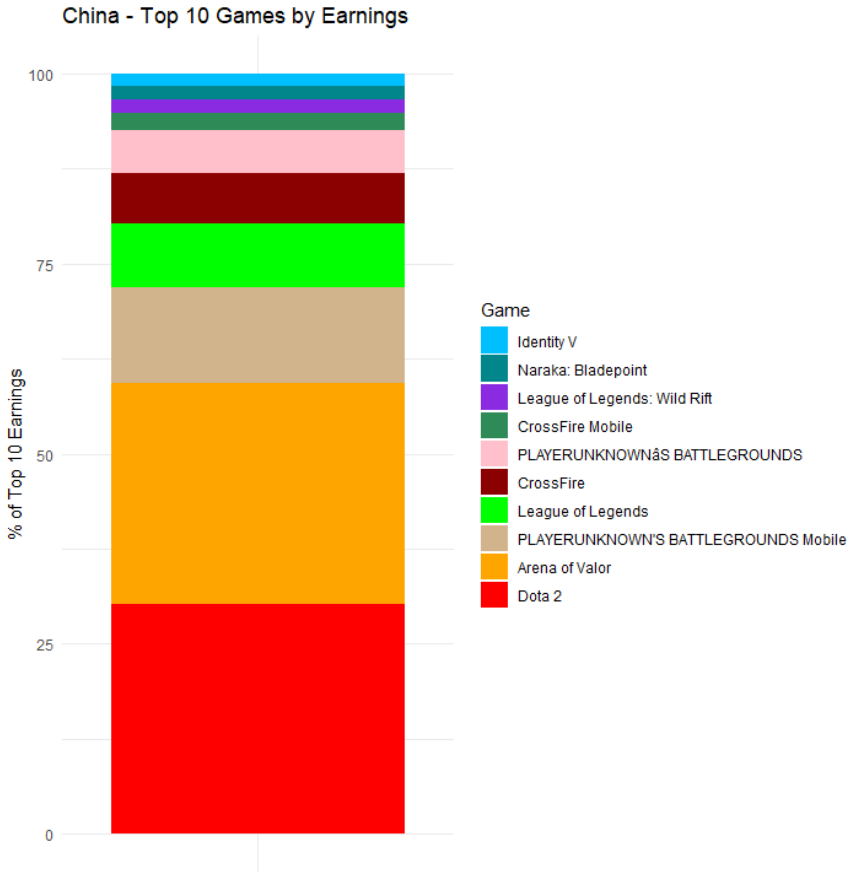


Figure 2

	Game	Pct
1	Fortnite	35.762415
2	Dota 2	13.102127
3	Rocket League	10.198863
4	Counter-Strike: Global Offensive	9.425263
5	Apex Legends	8.024147
6	Rainbow Six Siege	5.566971
7	Call of Duty: Warzone	5.163371
8	PLAYERUNKNOWNâ□□S BATTLEGROUNDS	4.922998
9	VALORANT	3.932004
10	Halo Infinite	3.901842

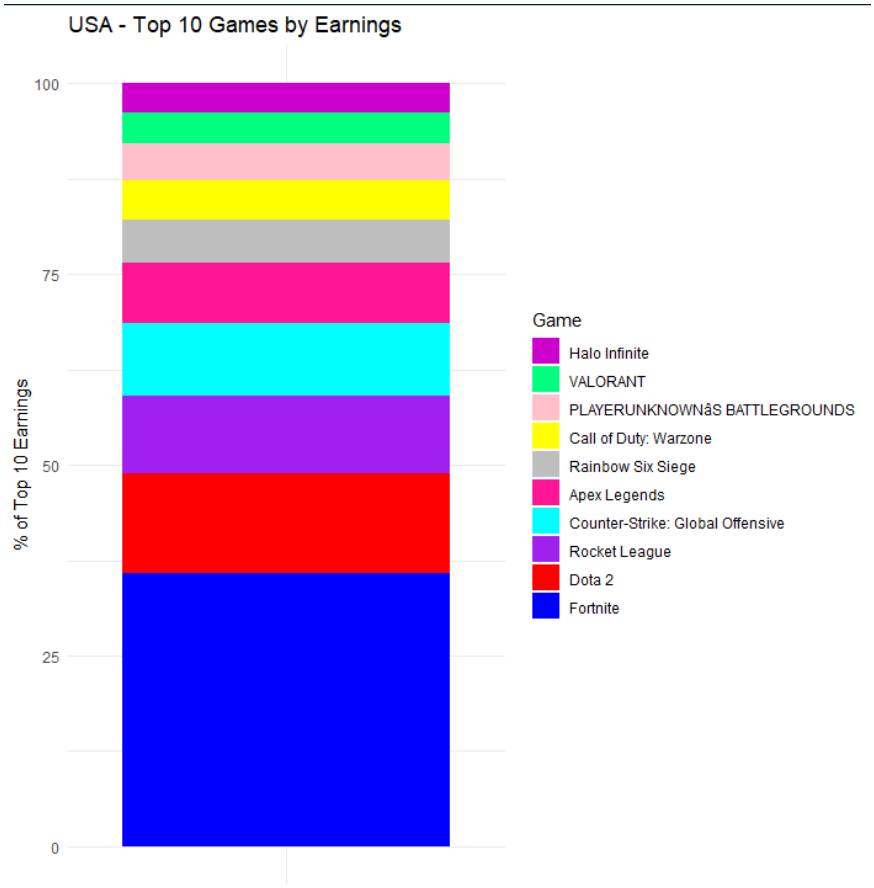


Figure 3

	Game	Pct
1	League of Legends	31.999626
2	StarCraft II	19.100242
3	Overwatch	13.854852
4	PLAYERUNKNOWNâ□□S BATTLEGROUNDS	9.722204
5	Overwatch 2	5.717763
6	StarCraft: Brood War	5.220286
7	StarCraft: Remastered	4.526990
8	Heroes of the Storm	4.495270
9	Dota 2	3.019037
10	VALORANT	2.343730

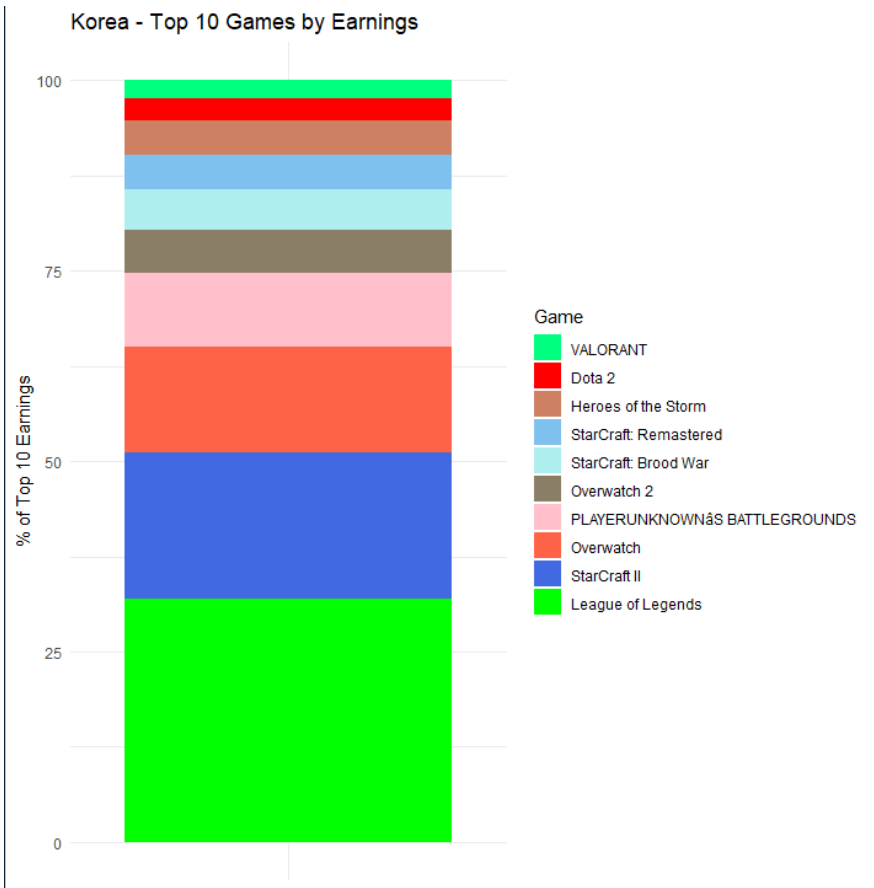


Figure 4

Alt Text: These stacked bar graphs show the differences in game preference across the three countries. In China (Figure 2), the eSport that has made the most money is Dota 2, which is responsible for over 30% of the total eSports earnings. Dota 2 also makes an appearance on

the other two countries top 10, coming second only to Fortnite in the United States. Fortnite, which only appears on the United States graph (Figure 3), makes up over 30% of the country's top 10 earnings. In Korea (Figure 4), the most popular game is League of legends, which also makes up over 30% of the country's total earnings. It's interesting how in each of these charts, the top game makes up a very similar percentage of the total earnings (around 30%). These graphs also show some intriguing geographical differences between the three countries. For example, in China, there are five mobile games in the top 10 (PUBG Mobile, Crossfire Mobile, Identity V, League of Legends Wild Rift, and Arena of Valor), which all together make up nearly 50% of the country's total eSports earnings. In contrast, neither the United States or Korea have any mobile games in their top 10. Lastly, there are only two games that appear in all three graphs, PUBG and Dota 2, indicating the global appeal of those games.

Author Contribution

Isabella A.: GitHub repository setup, Quarto document setup, data analysis 1, Plan document. Adam R.: Presentation setup, GitHub contributions, data analysis 2, data review. #Code Appendix

Citation

"Esports Earnings: Prize Money / Results / History / Statistics." Esports Earnings, http://www.esportsearnings.com. Accessed 28 Apr. 2026.